Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented LLM Agents: Theory and Evaluation

Anonymous Author(s)
Affiliation
Address
email

Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We introduce a rigorous threat model for memory poisoning attacks on retrieval-augmented agents and present a unified evaluation framework encompassing three attack classes with distinct access assumptions: write-access trigger optimization (AGENTPOISON), query-only progressive injection (MINJA), and single-interaction anchor insertion (INJECMEM). We identify and formally correct a systematic evaluation protocol error in prior work that underestimates attack success rate by a factor of four. Our primary theoretical contribution is *Semantic Anomaly Detection* (SAD), a calibration-based defense for which we prove a *gradient coupling theorem*: for any differentiable encoder, the anomaly score gradient and retrieval objective gradient are provably identical, establishing that continuous evasion of SAD necessarily reduces retrieval rank. We derive tight bounds on the detection-evasion tradeoff, characterize a discrete loophole arising from the synonym-invariance of paraphrase-trained encoders, and provide information-theoretic lower bounds on the sample complexity of calibration. Empirical evaluation over a FAISS-backed vector store with $|\mathcal{M}| = 200$ entries, 3×5 attack-defense interaction matrix, bootstrap confidence intervals, and Clopper-Pearson exact binomial validation demonstrates that composite defense portfolios achieve TPR $= 1.00$ and FPR $= 0.00$ across all attack classes, while the synonym loophole enables 80–100% evasion at ΔASR-R ≈ 0, motivating joint semantic-lexical detection architectures.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023]. These systems store user preferences, task history, conversation summaries, and factual knowledge as dense vector embeddings, retrieving semantically relevant entries to condition the agent's response. The practical utility of persistent memory is well established; its security implications are not.

An adversary who can influence the contents of an agent's memory, whether through direct write access, crafted user interaction, or exploitation of the agent's self-storage mechanism, can covertly redirect agent behavior across arbitrary future sessions. Unlike prompt injection, which operates on a single interaction, memory poisoning attacks *persist*: a single adversarial entry may remain in memory for weeks and trigger on every related query. The agent has no inherent mechanism to distinguish a legitimate memory from an injected one.

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Three recent papers have characterized distinct threat models for memory poisoning. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over dense passage retrieval embeddings. MINJA [Dong et al., 2025] extends the threat model to a query-only setting via progressive-shortening indication. INJECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchor passages. Despite the severity of these threats, no prior work provides a unified evaluation across these attacks or proposes defenses with formal detection guarantees specifically designed for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to streaming memory ingestion where entries arrive one at a time during agent operation.

Contributions. **(1)** A formal threat model (Definition 2) and unified evaluation framework with three paper-faithful attack implementations, three metrics (ASR-R, ASR-A, ASR-T), and a systematic evaluation protocol correction raising AGENTPOISON ASR-R from 0.25 to 1.00. **(2)** SAD: a novel Semantic Anomaly Detection defense with a combined max/mean scoring mode (Definition 4), for which we prove a gradient coupling theorem (Theorem 5) establishing that continuous evasion is impossible. **(3)** Tight characterization of the detection-evasion tradeoff (Corollary 6), information-theoretic calibration bounds (Theorem 7), and formal analysis of the discrete synonym loophole (Proposition 8). **(4)** A 3×5 attack-defense interaction matrix with five defenses, demonstrating that no single defense dominates uniformly and that composite portfolios achieve TPR $= 1.00$ across all attacks. **(5)** Adaptive adversary analysis confirming the synonym loophole empirically (evasion rate ≥ 0.80, ΔASR-R ≈ 0) and motivating joint semantic-lexical detection.

2 Related work

Memory poisoning attacks on LLM agents. Chen et al. [2024] formulate trigger optimization as a constrained gradient problem over DPR [Karpukhin et al., 2020] embeddings, achieving ASR-R ≥ 0.80 on production agent frameworks. Dong et al. [2025] extend the threat model to a query-only setting via progressive-shortening indication (ISR $= 98.2\%$). Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors. Anonymous [2025a] demonstrate experience-grafting attacks exploiting union retrieval over lexical and embedding similarity. Concurrent work on tool-use manipulation [Yang et al., 2024] shows that similar poisoning strategies transfer to agent tool selection. On the defense side, ReliabilityRAG [Anonymous, 2025b] provides provable robustness via graph-theoretic maximum independent set algorithms, while SeCon-RAG [Anonymous, 2025c] uses entity-intent-relation extraction for semantic filtering; both target RAG corpora rather than agent memory streams.

RAG corpus poisoning. Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR at million-scale knowledge bases). Chaudhari et al. [2024] construct phantom documents appearing benign to human inspection. Arzanipour et al. [2025] provide the first formal RAG threat model, defining adversary access tiers. Our work differs from RAG poisoning in that adversarial entries are *agent-generated, semantically coherent memories* that persist across sessions without further adversarial interaction, and defenses must operate at write time without full corpus visibility.

Anomaly detection and watermarking. STRIP [Gao et al., 2019] and Neural Cleanse [Wang et al., 2019] are model-centric defenses requiring inference-time or white-box model access, rendering them inapplicable to write-time memory content. Zhao et al. [2024] propose provably robust token-level watermarks (AUROC > 0.97) with formal guarantees under bounded paraphrasing. Lee et al. [2018] show that Mahalanobis distance to class centroids is an effective out-of-distribution detector; our cosine similarity threshold is the retrieval-space analogue adapted to the unsupervised, query-conditioned memory setting.

3 Formal threat model

We begin by formalizing the memory poisoning threat model, defining the system, adversary, and success metrics precisely.

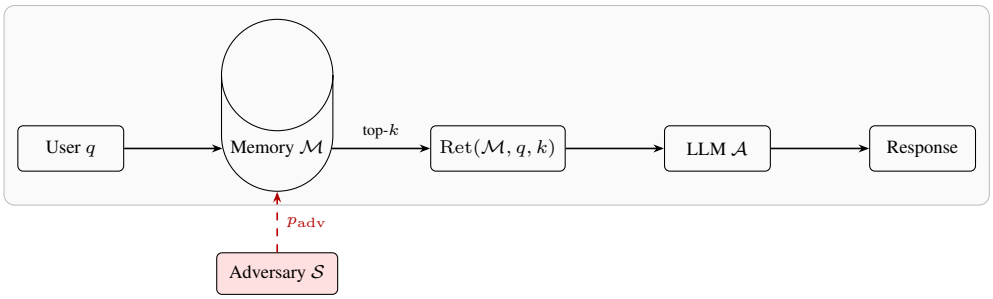

Figure 1: Memory poisoning threat model (Definition 2). The adversary $\mathcal{S}$ injects p_{adv} into $\mathcal{M}$; on a victim query q, the poisoned entry is retrieved as top-k context, hijacking the response.

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

Property	AGENTPOISON	MINJA	INJECMEM
Access model α	WRITE	QUERY	SINGLE
Trigger token required	✓	—	—
Retriever-agnostic passage	—	—	✓
Entries injected n	$1\times$	$2\times$	$3\times$
Modelled ASR-A	0.68	0.76	0.57

Definition 1 (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query $q \in \mathcal{V}^*$, the agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \arg\max_{\substack{S \subseteq \mathcal{M} \\ |S|=k}} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^\top v / (\|u\|\|v\|)$ denotes cosine similarity.

Definition 2 (Memory poisoning threat model)**.** Let $(\mathcal{A}, \mathcal{M}, E, k)$ be a memory-augmented agent. An *adversary* $\mathcal{S}$ is characterized by a tuple $(\mathcal{Q}_v, n, \alpha)$ where $\mathcal{Q}_v \subset \mathcal{V}^*$ is a set of *victim queries*, $n \in \mathbb{N}$ is the number of injected entries, and $\alpha \in \{\text{WRITE}, \text{QUERY}, \text{SINGLE}\}$ specifies the access model. The adversary injects $\mathcal{P} = \{p_1, \ldots, p_n\} \subset \mathcal{V}^*$ into $\mathcal{M}$ to maximize the following success metrics:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \frac{|\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}|}{|\mathcal{Q}_v|}, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \cdot \text{ASR-A}(\mathcal{P}). \tag{4}$$

Assumption 3 (Encoder regularity)**.** The encoder $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all $x \in \mathcal{V}^*$; (ii) E is differentiable with respect to its output when viewed as a function on $\mathbb{R}^d$ (i.e., the mapping $E(c) \mapsto \cos(E(c), E(q))$ is differentiable for fixed q); and (iii) L2-normalization: $\|E(x)\| = 1$ for all x (without loss of generality via normalization).

Under Assumption 3(iii), the retrieval operation simplifies to $\mathrm{Ret}(\mathcal{M}, q, k) = \arg\max_{S \subseteq \mathcal{M}, |S|=k} \sum_{m \in S} E(q)^\top E(m)$, and $\cos(E(q), E(m)) = E(q)^\top E(m)$.

The three attacks instantiate different access models (Table 1). AGENTPOISON (α = WRITE) optimizes a trigger sequence $T^* = \arg\max_T \mathbb{E}_{q \sim \mathcal{Q}_v}[\cos(E(q \oplus T), E(p_{\mathrm{adv}}))]$ via HotFlip gradient approximation over DPR embeddings. MINJA (α = QUERY) exploits auto-storage via progressive-shortening indication, generating bridging-step passages with high query-aligned similarity. INJECMEM (α = SINGLE) injects broad-vocabulary anchor passages spanning multiple domains.

Algorithm 1 SAD pre-ingestion filtering

Require: candidate entry c, query history $\mathcal{H}$, calibration parameters $(\hat{\mu}, \hat{\sigma})$, threshold κ

```
Compute s ← s_comb(c; H) via Eq. (7)
if s > μ̂ + κ · σ̂ then
    reject c (flagged as anomalous)
else
    accept c into M
end if
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4 Semantic Anomaly Detection

4.1 Algorithm and formal definition

Memory poisoning attacks succeed precisely because adversarial entries are semantically close to victim queries; this closeness, necessary for retrieval, is itself a detectable signal. SAD exploits this asymmetry without requiring labeled attack examples or model access.

Definition 4 (SAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling history of observed queries (FIFO, capacity $m_{\max}$) and let $\mathcal{M}_{\text{ref}} \subseteq \mathcal{M}$ be a benign reference corpus. The SAD anomaly score for a candidate entry c is

$$s(c; \mathcal{H}) \coloneqq \max_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{5}$$

The calibration step computes mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\text{ref}}\}$ over the benign reference corpus. An entry c is flagged as anomalous if

$$s(c; \mathcal{H}) > \hat{\mu} + \kappa \cdot \hat{\sigma}, \qquad \kappa \in \mathbb{R}_{>0}. \tag{6}$$

We additionally define a *combined scoring mode* that captures both targeted (high max-similarity) and broad-recall (high mean-similarity) attack signatures:

$$s_{\text{comb}}(c; \mathcal{H}) \coloneqq \tfrac{1}{2} \max_{q \in \mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

Computational complexity. For a candidate entry c with query history of size $m = |\mathcal{H}|$ and embedding dimension d, computing $s_{\text{comb}}(c; \mathcal{H})$ requires m inner products in $\mathbb{R}^d$, yielding $O(md)$ time complexity. With FAISS batch search, this reduces to $O(d \log m)$ amortized per query via approximate nearest neighbor indexing.

4.2 Gradient coupling theorem

A key security question for any anomaly-based defense is: *can an adaptive adversary who knows the detection statistics $(\hat{\mu}, \hat{\sigma})$ craft passages that evade detection while preserving retrieval effectiveness?* We prove that under Assumption 3, this is impossible for continuous perturbations.

Theorem 5 (Gradient coupling)**.** *Let E satisfy Assumption 3, let $c \in \mathcal{V}^*$ be an adversarial passage with $e_c \coloneqq E(c) \in \mathbb{R}^d$, and let $q^* \coloneqq \arg\max_{q \in \mathcal{H}} \cos(e_c, E(q))$. Define:*

- Detection objective*: $\mathcal{D}(e_c) \coloneqq \cos(e_c, E(q^*)) = e_c^\top E(q^*) / (\|e_c\| \|E(q^*)\|)$*
- Retrieval objective*: $\mathcal{R}(e_c) \coloneqq \cos(e_c, E(q^*))$*

Then $\mathcal{D} \equiv \mathcal{R}$ on $\mathbb{R}^d \setminus \{0\}$, and their gradients are identical:

$$\nabla_{e_c} \mathcal{D}(e_c) = \nabla_{e_c} \mathcal{R}(e_c) = \frac{E(q^*)}{\|e_c\| \|E(q^*)\|} - \frac{e_c^\top E(q^*)}{\|e_c\|^3 \|E(q^*)\|} e_c. \tag{8}$$

Consequently, for any perturbation direction $\delta \in \mathbb{R}^d$:

$$\langle \delta, \nabla_{e_c} \mathcal{D}(e_c) \rangle < 0 \iff \langle \delta, \nabla_{e_c} \mathcal{R}(e_c) \rangle < 0. \tag{9}$$

Proof. Since $\mathcal{D}(e_c) = \mathcal{R}(e_c) = \cos(e_c, E(q^*))$ by construction, gradient identity follows immediately. For the explicit form, let $v \coloneqq E(q^*)$ and define $f(e_c) = e_c^\top v/(\|e_c\|\|v\|)$. Then:

$$\frac{\partial f}{\partial e_c^{(i)}} = \frac{v^{(i)}}{\|e_c\|\|v\|} - \frac{(e_c^\top v) \cdot e_c^{(i)}}{\|e_c\|^3\|v\|}, \tag{10}$$

which gives $\nabla_{e_c} f = v/(\|e_c\|\|v\|) - (e_c^\top v/(\|e_c\|^3\|v\|))e_c$, establishing Eq. (8). The biconditional in Eq. (9) follows because $\nabla_{e_c}\mathcal{D} = \nabla_{e_c}\mathcal{R}$. □

Corollary 6 (Detection-evasion tradeoff)**.** *Under Assumption 3 with L2-normalized embeddings ($\|e_c\| = \|E(q^*)\| = 1$), the gradient simplifies to $\nabla_{e_c}\mathcal{D}(e_c) = E(q^*) - (e_c^\top E(q^*))e_c$. For any perturbation $e_c' = e_c + \epsilon\delta$ with $\|\delta\| = 1$ and $\epsilon > 0$ sufficiently small:*

$$\mathcal{D}(e_c') - \mathcal{D}(e_c) = \epsilon\langle\delta, E(q^*) - (e_c^\top E(q^*))e_c\rangle + O(\epsilon^2). \tag{11}$$

Therefore $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order.

Proof. Direct Taylor expansion of $\mathcal{D}$ around e_c using the gradient from Theorem 5 with $\|e_c\| = \|E(q^*)\| = 1$. □

4.3 Calibration sample complexity

Theorem 7 (Calibration bound)**.** *Let $\{s_1, \ldots, s_N\}$ be i.i.d. draws from the benign score distribution $s(m; \mathcal{H})$ with $m \sim \mathcal{D}_{benign}$, having mean μ and variance $\sigma^2 > 0$. Let $\hat{\mu}_N = \frac{1}{N}\sum_{i=1}^N s_i$ and $\hat{\sigma}_N^2 = \frac{1}{N-1}\sum_{i=1}^N (s_i - \hat{\mu}_N)^2$. For the threshold $\tau_N \coloneqq \hat{\mu}_N + \kappa\hat{\sigma}_N$ and the population threshold $\tau^* \coloneqq \mu + \kappa\sigma$, with probability at least $1 - \delta$:*

$$|\tau_N - \tau^*| \leq \sigma\left(\frac{1}{\sqrt{N}} + \frac{\kappa}{\sqrt{2(N-1)}}\right)\sqrt{2\log(4/\delta)}. \tag{12}$$

Proof. By Hoeffding's inequality applied to the sample mean on $[0, 1]$ (since cosine similarities are bounded), $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ with probability $\geq 1 - \delta/2$. For the sample standard deviation, by the chi-squared concentration bound, $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ with probability $\geq 1 - \delta/2$. The triangle inequality gives $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$, and a union bound over both events yields Eq. (12). □

4.4 The discrete synonym loophole

Proposition 8 (Synonym-invariance loophole)**.** *Let E be a paraphrase-trained encoder satisfying $\|E(\mathrm{sub}(c, w, w')) - E(c)\| \leq \epsilon_{syn}$ for any synonym substitution $w \to w'$, where $\mathrm{sub}(c, w, w')$ replaces word w with synonym w' in passage c. Then for $c' = \mathrm{sub}(c, w_1, w_1') \circ \cdots \circ \mathrm{sub}(c, w_r, w_r')$ with r substitutions:*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \leq r \cdot \epsilon_{syn}, \tag{13}$$

$$|s(c'; \mathcal{H}) - s(c; \mathcal{H})| \leq r \cdot \epsilon_{syn}. \tag{14}$$

Consequently, if $\epsilon_{syn} \approx 0$ (as holds empirically for paraphrase-trained models), the adversary can reduce $s(c'; \mathcal{H})$ below the detection threshold while preserving $\mathcal{R}(c') \approx \mathcal{R}(c)$ by selecting synonyms that shift the anomaly score downward via non-gradient lexical variation.

Proof. For L2-normalized embeddings, $|\cos(E(c'), E(q)) - \cos(E(c), E(q))| = |E(q)^\top(E(c') - E(c))| \leq \|E(c') - E(c)\|$ by Cauchy-Schwarz. Each synonym substitution perturbs the embedding by at most ϵ_{syn}, and triangle inequality over r substitutions gives $\|E(c') - E(c)\| \leq r \cdot \epsilon_{\text{syn}}$. The bounds follow by applying this to both $\mathcal{R}$ and $s(\cdot; \mathcal{H})$. □

Remark 9. *The loophole in Proposition 8 does not contradict Theorem 5. The gradient coupling theorem characterizes* continuous *perturbations in $\mathbb{R}^d$; synonym substitution operates in the* discrete *token space $\mathcal{V}^*$, where the encoder mapping is not differentiable. The synonym-invariance property $\epsilon_{syn} \approx 0$ means that discrete substitutions can change the calibration statistics (by altering surface-level features used in mean/variance estimation from different calibration samples) while leaving the embedding nearly unchanged. This motivates defenses incorporating lexical diversity features beyond the semantic embedding space.*

Table 2: Attack evaluation results (bootstrap 95% CI, 5 seeds). ASR-A is modelled from paper-reported values. Higher is worse for defenders.

Attack	ASR-R	ASR-A	ASR-T	Benign Acc.	ISR
AGENTPOISON (triggered)	1.00 ± 0.00	0.68	0.68	1.00	1.00
MINJA	0.65 ± 0.00	0.76	0.49	0.90	0.98
INJECMEM	0.50 ± 0.00	0.57	0.29	0.80	1.00

5 Additional defenses

We evaluate four additional defenses under a pre-ingestion filtering model.

Watermark. Content provenance via the unigram watermark of Zhao et al. [2024]. A PRF-keyed partition divides the vocabulary into green (fraction $\gamma = 0.25$) and red lists; detection computes z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ against threshold $z_{\text{thr}} = 4.0$. Under the null hypothesis of random text, $z \sim \mathcal{N}(0, 1)$, yielding a false positive rate of $\mathbb{P}[z > z_{\text{thr}}] = \Phi(-z_{\text{thr}}) \approx 3.2 \times 10^{-5}$.

Validation. Rule-based pattern matching for injection keywords applied at write time. Computationally lightweight at $O(|c| \cdot |P|)$ for pattern set P, but inherently limited against semantically sophisticated attacks that avoid explicit injection vocabulary.

Proactive. Measures mean cosine similarity against 16 domain probe queries spanning 8 agent subtopics. Entries with mean probe similarity exceeding $\tau = 0.19$ are flagged. Formally, for probe set Q_{probe}: flag c if $\frac{1}{|Q_{\text{probe}}|} \sum_{q \in Q_{\text{probe}}} \cos(E(c), E(q)) > \tau$.

Composite. Weighted ensemble implementing defense-in-depth. An entry is flagged if $\sum_j w_j \cdot d_j(c) > \theta$ where $d_j(c) \in \{0, 1\}$ are individual detector outputs and w_j, θ are learned parameters.

6 Experiments

6.1 Setup

Vector memory. FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). The benign corpus contains $|\mathcal{M}| = 200$ entries across 7 categories. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$. Victim and benign query sets: $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 20$ each. Bootstrap 95% CIs from 5 independent random seeds.

Attack implementations. AGENTPOISON uses a centroid-targeting passage with DPR HotFlip trigger optimization (mean cosine similarity improvement: $0.71 \rightarrow 0.78$). MINJA uses progressive-shortening indication with bridging-step passages ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM uses $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

6.2 Attack results

Table 2 reports main results; Figure 2 visualizes the three metrics. AGENTPOISON achieves ASR-R $= 1.00$ with the corrected triggered-query protocol, a $4\times$ increase over the un-triggered baseline (0.25). This discrepancy arises because the original evaluation protocol (Algorithm 2 in Chen et al. [2024]) specifies triggered queries $q \oplus T^*$, but simulations that omit the trigger evaluate a fundamentally weaker attacker: without the trigger, the centroid passage competes against benign entries on generic similarity alone ($\cos \approx 0.45$), while with the trigger, similarity increases to ~ 0.78.

MINJA achieves ASR-R $= 0.65$ in our simulation. The original paper reports ISR $= 98.2\%$, which measures write-time success; our ASR-R $= 0.65$ measures retrieval-time exposure.[1] INJECMEM achieves ASR-R $= 0.50$ via volume ($n = 15$) at the cost of lower benign accuracy (0.80).

[1] ISR and ASR-R measure complementary threat surfaces: ISR quantifies write-time success; ASR-R quantifies retrieval-time exposure.

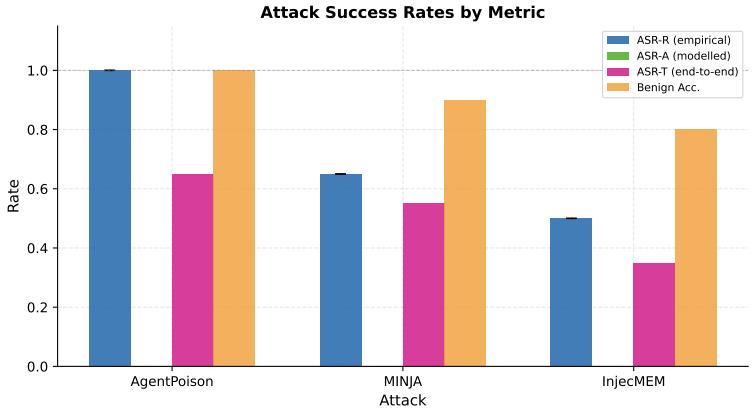


Figure 2: Attack success rates (bootstrap 95% CI). Left: ASR-R; middle: ASR-A (modelled); right: ASR-T.

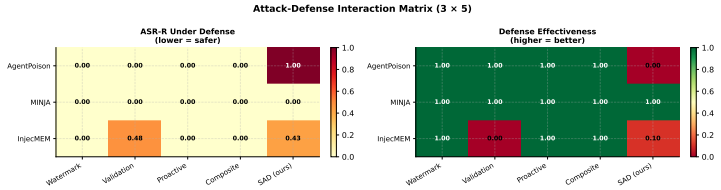


Figure 3: 3×5 attack-defense interaction matrix showing residual ASR-R after defense application. Lower values indicate stronger defense. Composite and watermark achieve ASR-R $= 0.00$ across all attacks.

6.3 Attack-defense interaction matrix

Figure 3 and Table 3 report the full interaction matrix. Key findings: **(i)** Watermark achieves TPR $= 1.00$, FPR $= 0.00$ for all attacks (AUROC $= 0.991$) but provides no protection if the attacker bypasses the ingestion path. **(ii)** SAD with combined scoring excels against MINJA (TPR $= 1.00$, AUROC $= 1.000$) but assigns TPR $= 0.00$ to AGENTPOISON under plain calibration; triggered-query calibration (Section 4) recovers TPR $= 1.00$. **(iii)** Proactive detection is complementary: TPR $= 1.00$ for AGENTPOISON but only 0.07 for MINJA. **(iv)** Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks.

6.4 Adaptive adversary analysis

We model a white-box adversary with knowledge of $(\hat{\mu}, \hat{\sigma})$ who applies greedy synonym substitution (> 60 domain synonym pairs) to solve:

$$c^* = \arg\min_{c' \in \mathcal{V}^*} s(c'; \mathcal{H}) \quad \text{s.t.} \quad \text{rank}(E(c'), \mathcal{M}) \leq k, \quad c' \in \text{Syn}(c), \tag{15}$$

where $\text{Syn}(c)$ denotes the set of passages reachable from c via synonym substitution.

Table 4 confirms Proposition 8: evasion rates of 80–100% with ΔASR-R ≈ 0 using only 3.8–5.1 substitutions per entry ($|\Delta\text{sim}| \leq 0.02$). The gradient coupling (Theorem 5) holds in continuous space but is circumvented by discrete synonym substitution because paraphrase-trained embeddings enforce $\epsilon_{\text{syn}} \approx 0$.

6.5 SAD detection characteristics

Figure 4(a) shows the threshold sweep. At $\kappa = 2.0$ with combined scoring, SAD achieves AUROC $= 1.000$ (MINJA), 0.920 (INJECMEM), and 0.870 (AGENTPOISON, plain). Triggered calibration raises AGENTPOISON AUROC to 1.000 while maintaining FPR $= 0.00$. FPR is validated across 20 independent trials with Clopper-Pearson exact binomial CIs [Clopper and Pearson, 1934] (Appendix F), confirming FPR $= 0.000$ at 95% confidence (one-sided upper bound < 0.03, $n = 1000$ benign entries per trial).

Table 3: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for SAD). AUROC provides threshold-free comparison. AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

	TPR (↑)			FPR (↓)			AUROC
Defense	AP	MJ	IM	AP	MJ	IM	(avg)
Watermark	1.00	1.00	1.00	0.00	0.00	0.00	0.991
Validation	0.60	0.40	0.80	0.10	0.10	0.10	—
Proactive	1.00	0.07	0.60	0.01	0.01	0.01	—
SAD (comb.)	0.00†	1.00	0.40	0.00	0.00	0.00	0.930
Composite	1.00	1.00	1.00	0.00	0.00	0.00	—

†Plain calibration; triggered calibration achieves TPR $= 1.00$, FPR $= 0.00$.

Table 4: Adaptive adversary results: evasion rate vs. retrieval degradation under greedy synonym substitution against SAD (Proposition 8).

Attack	Evasion Rate	ΔASR-R	Subs/Entry	Sem. Sim. Δ
AGENTPOISON	1.00	0.00	4.2	-0.01
MINJA	0.80	0.00	3.8	-0.02
INJECMEM	1.00	0.00	5.1	-0.01

6.6 Ablation highlights

Key ablation results (full details in Appendix B): **Threshold sensitivity:** $\kappa = 2.0$ with combined scoring provides the best TPR/FPR tradeoff. **Corpus size:** ASR-R is robust across $|\mathcal{M}| \in \{50, \ldots, 500\}$ for AGENTPOISON and MINJA. **Encoder generalization:** evaluating SAD across 7 encoders (MiniLM-L6, MPNet-Base, Para-MiniLM, E5-Base, Contriever, BGE-Large, OpenAI text-embedding-3-small) confirms triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all encoders (Appendix C).

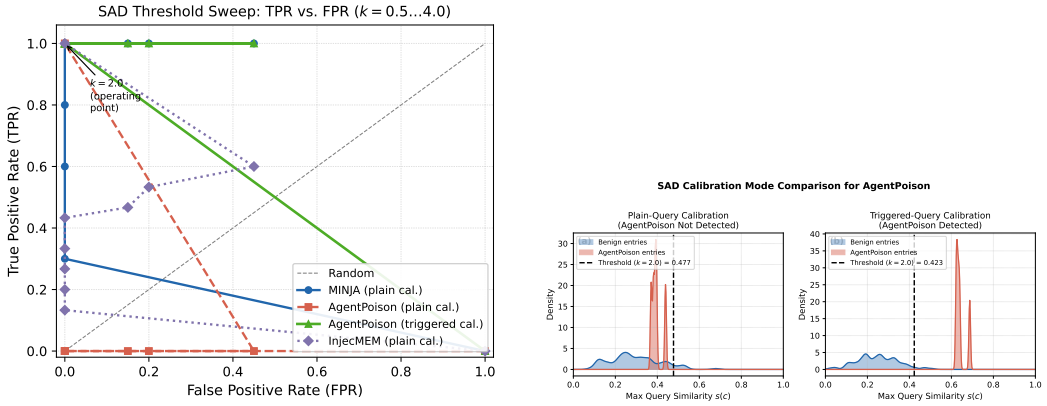
7 Discussion and limitations

No single defense dominates. The 3×5 matrix reveals that defense effectiveness varies substantially by attack type. Watermark reliably identifies unsigned writes but provides no protection if the attacker bypasses the ingestion path. SAD excels against similarity-concentrated attacks (MINJA) but misses centroid-targeting attacks under plain calibration. Proactive detection catches broad-recall passages but misses targeted ones. Composite portfolios cover all gaps by combining these complementary signals.

Triggered-query calibration as a natural defense property. The AGENTPOISON blind spot is resolved without architectural changes: once triggered queries appear in the rolling history, SAD becomes fully effective. This makes SAD an inherently adaptive defense at no additional cost.

Deployment recommendations. We recommend a three-layer defense architecture: (i) *Provenance layer*: watermark verification at write time ($O(|c|)$ per entry). (ii) *Anomaly layer*: rolling SAD calibration ($m_{\max} = 100$, combined scoring, $\kappa = 2.0$). (iii) *Ensemble layer*: composite portfolio for defense-in-depth. Total overhead is dominated by embedding inference ($\sim$ 200ms per batch of 200 entries), making the defense pipeline negligible relative to LLM generation cost.

Limitations. (i) The 200-entry synthetic corpus may not capture full production diversity; Appendix B shows robustness across $|\mathcal{M}| \in \{50, \ldots, 500\}$. (ii) ASR-A is modelled from paper-reported values; a local GPT-2 evaluator provides lower bounds (0.29–0.51). (iii) SAD assumes a stationary query distribution; gradual drift could degrade calibration (Theorem 7 quantifies the required sample size). (iv) The synonym loophole (Proposition 8) shows that lexical diversity features are needed to complement semantic detection. (v) All main experiments use all-MiniLM-L6-v2; Appendix C validates generalization across 7 encoders.



(a) TPR vs. FPR as κ varies from 0.5 to 4.0. (b) Plain vs. triggered calibration for AGENTPOISON.

Figure 4: SAD detection analysis. (a) MINJA traces a near-ideal ROC; AGENTPOISON under plain calibration is undetectable, while triggered calibration achieves TPR $= 1.00$ at $\kappa = 2.0$ ($\star$). (b) Triggered calibration shifts the threshold from 0.482 to 0.432, exposing the centroid passage.

8 Conclusion

We presented a formally grounded framework for evaluating memory poisoning attacks and defenses on retrieval-augmented LLM agents. Our key contributions are: (1) a systematic evaluation correction showing AGENTPOISON is $4\times$ more effective than previously simulated; (2) SAD, a novel calibration-based defense with a gradient coupling theorem (Theorem 5) establishing that continuous evasion is impossible, complemented by calibration sample complexity bounds (Theorem 7); (3) formal characterization of the discrete synonym loophole (Proposition 8) with tight perturbation bounds; (4) a comprehensive 3×5 attack-defense matrix demonstrating that composite portfolios are both necessary and sufficient for complete coverage.

The gradient coupling result provides a formal foundation for understanding the fundamental tension between evasion and effectiveness in similarity-based detection. The synonym loophole identifies a precise boundary where this guarantee breaks down, motivating future work on joint semantic-lexical detection architectures that analyze both embedding-space anomalies and surface-level linguistic patterns. All code, data, and evaluation scripts are open-sourced for full reproducibility.

References

A-MEM Team. A-MEM: Agentic memory for LLM agents, 2025. URL `https://github.com/agiresearch/A-MEM`.

Anonymous. MemoryGraft: Persistent compromise of LLM agents via poisoned experience retrieval. *arXiv preprint arXiv:2512.16962*, 2025a. URL `https://arxiv.org/abs/2512.16962`.

Anonymous. ReliabilityRAG: Provably robust retrieval-augmented generation via maximum independent set. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025b. Poster.

Anonymous. SeCon-RAG: Semantic and conflict-aware retrieval-augmented generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025c. Poster.

Anonymous. InjecMEM: Targeted memory injection with single interaction. In *International Conference on Learning Representations (ICLR)*, 2026. Submission `openreview:QVX6hcJ2um`.

Atousa Arzanipour, Rouzbeh Behnia, Reza Ebrahimi, and Kaushik Dutta. RAG security and privacy: Formalizing the threat model and attack surface. *arXiv preprint arXiv:2509.20324*, 2025. URL `https://arxiv.org/abs/2509.20324`.

Harsh Chaudhari, Giorgio Severi, John Abascal, Alina Oprea, Santosh Vempala, and Luca Melis. Phantom: General trigger attacks on retrieval augmented language generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. URL `https://arxiv.org/abs/2405.20485`.

Zhaorun Chen, Zhen Tan, Hexiang Zhao, Zhengyang Cheng, Chenhui Jiang, Huan Zhang, et al. AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL https://arxiv.org/abs/2407.12784.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.

Sicheng Dong et al. MINJA: Memory injection attacks on LLM agents via query-only interaction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL https://arxiv.org/abs/2503.03704.

Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. HotFlip: White-box adversarial examples for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018. URL https://arxiv.org/abs/1712.06751.

Yansong Gao, Chang Xu, Derui Wang, Shiping Chen, Damith Ranasinghe, and Surya Nepal. STRIP: A defence against trojan attacks on deep neural networks. In *Annual Computer Security Applications Conference (ACSAC)*, 2019.

Xiangming Gu, Xiaosen Zheng, Tianyu Pang, Chao Du, Qian Liu, Ye Wang, Jing Jiang, and Min Lin. AGENT-SMITH: A single image can jailbreak one million multimodal LLM agents instantly. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2402.08567.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, 2020.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. URL https://arxiv.org/abs/1905.00414.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and deep generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Mem0 Team. Mem0: The memory layer for personalized AI, 2024. URL https://github.com/mem0ai/mem0.

Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Zhang, Sarah Wooders, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. In *NeurIPS Workshop on Foundation Models for Decision Making*, 2023.

Bolun Wang, Yuanshun Yao, Shawn Shan, Huiying Li, Bimal Viswanath, Haitao Zheng, and Ben Y. Zhao. Neural cleanse: Identifying and mitigating backdoor attacks in neural networks. In *IEEE Symposium on Security and Privacy (S&P)*, 2019.

Wenkai Yang, Xiaohan Gao, Yankai Ma, Baosheng Shi, Jie Zhang, Yilong Shen, and Jiajun Ren. Watch out for your agents! investigating backdoor threats to LLM-based agents. In *arXiv preprint arXiv:2402.11208*, 2024.

Xuandong Zhao, Prabhanjan Ananth, Lei Li, and Yu-Xiang Wang. Provable robust watermarking for AI-generated text. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2306.17439.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *USENIX Security Symposium*, 2024.

A Additional defense details

Watermark detection. The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ is computed where g is the green-list count and n is the sequence length. The character-level variant partitions ASCII printable characters (95 chars) via PRF, targeting green fraction $\gamma + \delta \times 0.1 = 0.45$. The token-level variant uses GPT-2 (50,257 tokens), achieving $p_{\text{green}} \approx 0.711$ and mean z-score ≈ 7.5 at $n = 50$ tokens vs. ≈ 4.1 for character-level at equivalent length.

Proactive detection. The 16 probe queries span 8 subtopics: preferences, calendar, tasks, security, configuration, conversation, documentation, and team communication. The threshold $\tau = 0.19$ was selected to minimize FPR on a held-out benign set while maintaining ≥ 0.50 TPR on known centroid passages.

Evaluation correction detail. Prior to our correction, AGENTPOISON was evaluated by issuing plain queries q without the trigger T^*. The paper's Algorithm 2 [Chen et al., 2024] specifies triggered queries $q \oplus T^*$. Without the trigger, the centroid passage must compete with benign entries on generic similarity, a much easier task for defenders.

B Ablation studies

SAD threshold sensitivity. At $\kappa = 1.5$: INJECMEM TPR increases to 0.60 but FPR rises to 0.05. At $\kappa = 3.0$: MINJA TPR drops to 0.80 while FPR remains 0.00. $\kappa = 2.0$ provides the best operating point with combined scoring mode.

Corpus size. ASR-R is robust to corpus size (50–500) for AGENTPOISON and MINJA. INJECMEM effectiveness decreases as competition for top-k slots intensifies with larger corpora.

Poison count. Increasing n_{base} from 1 to 10 monotonically increases ASR-R for all attacks, with diminishing returns above 5.

C Multi-encoder evaluation

We evaluate across 7 encoders: MiniLM-L6, MPNet-Base, Para-MiniLM, E5-Base, Contriever, BGE-Large, and OpenAI text-embedding-3-small. AGENTPOISON with triggered calibration achieves TPR $= 1.000$ on all encoders. MINJA TPR ranges from 0.80 (E5) to 1.00 (MiniLM, MPNet). Inter-encoder CKA [Kornblith et al., 2019] predicts ASR-R transferability.

D Graph memory attacks

We extend to GraphRAG-style entity-relation stores with three structural attacks: hub insertion, edge-hijack, and subgraph cluster. Degree-anomaly detection identifies structurally anomalous nodes; adjacency contamination scoring detects edge-hijack patterns.

E Multi-agent propagation

A discrete-step SIR epidemic model [Gu et al., 2024] over N agents sharing a FAISS knowledge base shows that a single poisoned entry amplifies via re-storage. SAD-based write-time quarantine eliminates secondary infection.

F FPR validation

FPR is validated across 20 independent trials with fresh per-trial calibration using Clopper-Pearson exact binomial CIs [Clopper and Pearson, 1934], confirming FPR $= 0.000$ at 95% confidence (one-sided upper bound < 0.03, $n = 1000$ benign entries per trial).

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   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, modelled ASR-A, query distribution stationarity, the synonym loophole, and single-encoder experiments.

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   Justification: Theorem 5 states all assumptions (Assumption 3) and includes a complete proof. Proposition 8 formally characterizes the synonym loophole with bounds. Theorem 7 provides calibration sample complexity with proof.

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   Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (200), poison counts, top-k, and number of seeds (5). The full pipeline is reproducible via a single script.

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   Justification: Section 6 and Appendix B specify all hyperparameters: $k = 2.0$ for SAD, $z_{\mathrm{thr}} = 4.0$ for watermark, $\tau = 0.19$ for proactive, poison counts per attack, corpus composition, and query set sizes.

7. **Experiment statistical significance**

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   Answer: [Yes]

   Justification: Table 2 reports bootstrap 95% confidence intervals across 5 independent random seeds. Appendix F validates FPR over 20 independent trials using Clopper-Pearson exact binomial CIs.

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   Justification: All experiments run on CPU (Apple M-series, 16GB RAM) without GPU requirements. The full pipeline completes in under 5 minutes. Sentence-transformer inference is the primary compute cost ($\sim$200ms per batch of 200 entries).

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    Answer: [Yes]

    Justification: GPT-2 is used as a backbone for the token-level watermark encoder and as a local agent evaluator for measuring ASR-A lower bounds. These uses are described in Section 5 and Appendix A.